# SE(3)-WAM: What Should a Robot Remember and Predict for Dynamic Manipulation?

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## Abstract

*Dynamic manipulation requires contact with a scene that changes between observation and execution. What should a robot remember and predict to act at the right place and time? We introduce SE(3)-WAM, a structured world–action model and a controlled study of this question. The model connects timestamped history, metric target dynamics, scene prediction, and flow-matching action generation. Its design separates three factors: the motion information retained from history, the variables and resolution of future prediction, and the information exchanged between world and action streams. A fourteen-task benchmark evaluates these choices through task success, contact geometry, timing, and computation cost. Diagnostic comparisons show that readable history and action sensitivity need not yield better control, and that future-supervision benefits depend on the surrounding representation. Matched component studies and forecast interventions distinguish these effects from useful online anticipation. The training procedure builds a robot-data base that independently initializes simulation and real-robot adaptation; the system evaluation covers dynamic transfer, static compatibility, and measured execution delay. Together, the model and protocol connect prediction quality to the information and time available for action.*

## 1. Introduction

A robot reaching for a moving object must arrive where the object will be, with the orientation and timing required for contact. The same requirement appears when releasing into a moving receptacle or passing through an opening that is closing. A current image can localize the scene yet leave its motion ambiguous; computation and execution then make that image older. Dynamic manipulation therefore calls for both temporal information and actions that remain appropriate when applied [9, 22, 34].

What should a robot remember and predict? Reconstructing every visible change is one answer, but the control problem does not assign equal importance to every pixel. Target pose determines contact geometry. An obstacle matters through reachability and collision, an occluder through visibility, and static geometry can remain essential despite having no motion. Our organizing principle is to preserve motion and context at the resolution required by interaction. A fixed target/context decomposition and a shared scene representation are two testable implementations of this principle.

World–action models learn future scene variables alongside robot actions [37, 38]. They offer several ways to implement this idea, but also combine distinct sources of improvement: extra supervision, a different state representation, and additional computation before acting. An auxiliary future loss can shape useful features without an online forecast [40]; a generated future can instead serve as an explicit input to action. Comparing these mechanisms requires separating what the policy observes, what it learns to predict, and what it consumes at deployment.

We introduce SE(3)-WAM, a structured world–action model and design study for dynamic manipulation. Its timestamped history encoder supplies a common conditioning bank; a world branch represents target motion and scene evolution; an action branch generates commands in a metric robot frame. Three questions organize the design. **Q1** concerns the information retained from history. **Q2** concerns the scope, variables, and resolution of future prediction. **Q3** concerns parameter sharing, world–action access, and generation order. The study compares these choices under controlled interfaces and checks their interactions before selecting a complete model.

This separation is consequential even in the diagnostic comparisons. Structured motion history improves a reference policy, whereas more readable visual history does not necessarily improve control. Future supervision is also recipe dependent: sparse trajectories yield little overall change under one reference, but a larger, task-concentrated benefit with another encoder and reconstruction objective. These findings motivate evaluating the complete information path, rather than assuming that more history or more

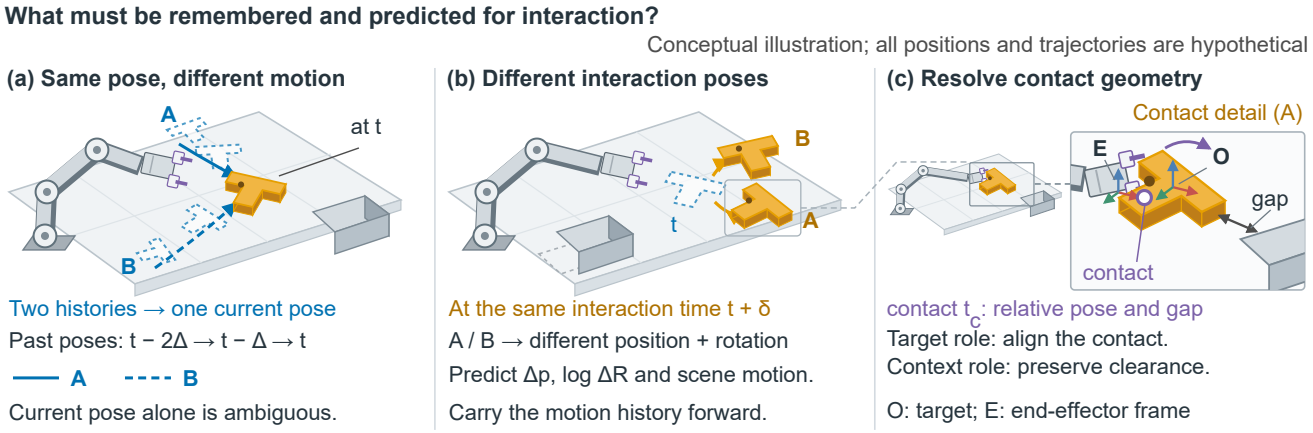



Figure 1. Dynamic manipulation requires information about interaction time. Two histories reach the same current target pose (a), but imply different poses at a common future instant (b). Successful contact also depends on orientation and surrounding clearance (c). Solid and dashed trajectories denote the two conceptual motions, not measured rollouts.

prediction is sufficient. The full evaluation combines these diagnostics with matched design comparisons, real-robot-data pretraining, and independent simulation and real-robot adaptation.

Our contributions are:

- A fourteen-task dynamic manipulation benchmark and evaluation protocol that connect task success to motion, contact, and observation-to-action delay.
- A controlled study of history, future representation, and world–action coupling, with interventions that distinguish information retention, action dependence, and closed-loop utility.
- A complete SE(3)-WAM model family and training procedure, with structured target dynamics, configurable scene prediction, and a common robot-data pretraining base for independent downstream adaptation.

## 2. Related Work

**Dynamic manipulation and physical time.** Motion-aware grasping connects prediction to reachability and interception [1]. DOMINO/PUMA couples a dynamic benchmark with historical flow and object-future supervision [9]; DynamicManip expands static demonstrations into dynamic data [16]. DynamicVLA emphasizes continuous inference and action streaming [34], while DynamicWAM conditions world–action learning on motion [22]. Temporal feature prediction [43], AHEAD's predictive latents [27], and ForeTime-VLA's future-token and event-time supervision [23] offer different ways to anticipate interaction. RTC addresses continuity under delayed chunk generation [3]; Chronos Policy explicitly represents observation and execution times [29]. These works motivate evaluating prediction together with action age, rather than identifying a simulator control interval with real-time operation.

**History and object-centered information.** TraceVLA encodes temporal traces as visual prompts [46]; HiF-VLA organizes past and future dynamics through motion representations [17]. PACT-WAM allocates fewer spatial tokens to older observations [20], and Long-WAM studies long context together with causal video pretraining [12]. Thus a history encoder's value depends on both temporal support and the training that makes that information usable. PointWorld explicitly assumes a static initial scene when neither prior frames nor velocities are supplied [11]. Externally moving targets make this initial-state information a central factor in Q1. DOCIR separates manipulated objects, obstacles, and the robot [8]; OA-WAM makes objects addressable within world–action modeling [21]. Our Q1 comparisons distinguish current localization, privileged motion, and visual history before comparing encodings. Role assignment is an experimental factor, not a presumed optimum.

**Which future should a policy predict?** SE3-Nets models scene motion through rigid transforms [4]; SE3-Pose-Nets uses structured pose dynamics for control [5]. DSR-Net accumulates a volumetric object representation over interactions [36]; PhysMani learns 3-D world representations for manipulation [41]. ATM learns point trajectories from video [32]; Im2Flow2Act uses generated object flow as a policy interface [35]. AffordTrajDP propagates contact affordances with object pose [33], and FlowWAM uses flow in policy and world-model modes [6]. These interfaces differ in whether they describe desired motion, action-conditioned consequences, or externally moving objects. DreamVLA and DreamWAM add complementary visual, geometric, and semantic targets [39, 42]; DC-WAM instead emphasizes

dynamics within RGB supervision [13]. DELE-w0.5 uses compact future latents [15]. Q2 places related target families under one action path to measure their conditional training value; its pose output does not imply a new rigid-motion parameterization.

**World–action coupling and experimental attribution.** Prediction enters control through different interfaces. PointWorld predicts action-conditioned 3-D point flows for candidate-action evaluation [11]; NWM simulates and ranks navigation trajectories [2]. CoT-VLA generates a visual goal before its action sequence [44], while VPP conditions actions on predictive features [10]. DreamZero jointly models world and action [37]; Fast-WAM separates training benefits from test-time imagination [40]. Our reference D uses future supervision without a forecast-to-action edge. Q3 changes this access and generation order under controlled inputs. Wang et al.'s OpenWAM supplies the Alpha foundation for our adapted baseline [30]; Yu et al.'s OpenWAM studies composable world–action programs [38]. ACT, Diffusion Policy, and $\pi_{0.5}$ supply system references [7, 24, 45]; DROID supplies real-robot pretraining trajectories [14]. Native system comparisons and matched design contrasts answer different questions and are reported separately.

## 3. Dynamic Manipulation Benchmark

**Interaction in physical time.** Let $\mathcal{H}_t = \{(o_i, s_i, \tau_i)\}_{i \le t}$ contain the available finite history of observations, robot states, and timestamps. A policy outputs $A_t = (a_{t,0}, \ldots, a_{t,H_a-1})$. For image capture time $\tau_t^s$, observation-to-command delay $\ell_t$, and action interval $\Delta_a$,

$$\tau_{t,j}^{\text{apply}} = \tau_t^s + \ell_t + j\Delta_a. \tag{1}$$

Prediction horizons are measured from capture time; contact is assessed at the actual application time. This distinguishes anticipating motion from merely increasing the number of future action tokens.

**Tasks and demonstrations.** Our MuJoCo/robosuite suite [28, 47] contains fourteen tasks covering moving-target and orientation-sensitive grasping, physical occlusion, moving-receptacle placement, phase-constrained passage, rolling/sliding objects, and post-impact adaptation. The DB0 v0.6 demonstration pool contains 11,200 trajectories, 800 per task. Native evaluators measure physical task completion, with equal weight per task. Appendix A specifies motions, event conditions, thresholds, and data construction; recorded observations appear in Supplementary Fig. S1.

**Information and execution tracks.** The *visual deployment track* uses causal sensor observations and robot state, including estimated target motion where required by the encoder. The *current-pose track* additionally supplies exact current target pose; the *object-motion track* supplies its interval kinematics as well. These oracle tracks separate representation quality from localization and tracking errors. The diagnostic screens use synchronous replanning with one executed command; native command delay is 50 ms for T5 and zero otherwise. The deployment evaluation advances the scene during measured inference and reports a separate imposed-delay sweep. Thus synchronous scores and wall-clock robustness occupy different columns.

**Splits and scores.** The existing fixed140 partition (ten scenes per task) and its 1,260-scene complement form a development pool. Their identities are preserved in all diagnostic results. Final confirmation uses a separate 1,400-scene set, with new scene seeds, no training/development episodes, and the same task generator. Model selection uses development scenes only. Three independent training seeds share evaluation scenes. For binary outcomes $y_{mki}$,

$$\mathrm{SR}(m) = \frac{1}{14}\sum_{k=1}^{14}\frac{1}{n_k}\sum_{i=1}^{n_k} y_{mki}. \tag{2}$$

We report seed-wise scores and paired differences; the confirmation intervals resample training seeds and scenes as separate factors. The earlier single-seed screens have scene-only intervals. Contact-time and relative-pose errors, collision/failure categories, action age, and p95 latency explain the task score. Appendix A defines their support and the treatment of episodes without contact.

## 4. SE(3)-WAM

We represent a model by $(H, Z, B, G)$: its history encoder, future representation, backbone, and world–action computation program. Figure 2 shows the complete interface. Future observations provide training supervision; at deployment, all configurations condition on observed history and generated states. A concrete reference uses eight RGB observations with target-motion history, target pose and scene trajectories as future variables, a shared-attention backbone, and auxiliary world supervision. Q1–Q3 vary these choices while preserving the relevant input and execution contracts.

### 4.1. Timestamped History and Scene Roles

The history encoder produces $X_t = H(\mathcal{H}_t)$. Each observation token combines visual content, physical age $\tau_t^s - \tau_i$, and geometric information when available. Robot-state tokens encode end-effector pose and measured gripper opening. Object-history tokens retain metric positions, relative rotations, interval lengths, and visibility. Causal estimates use only frames through $t$; offline smoothing is restricted to supervision. This distinction also applies when training on real-robot trajectories.

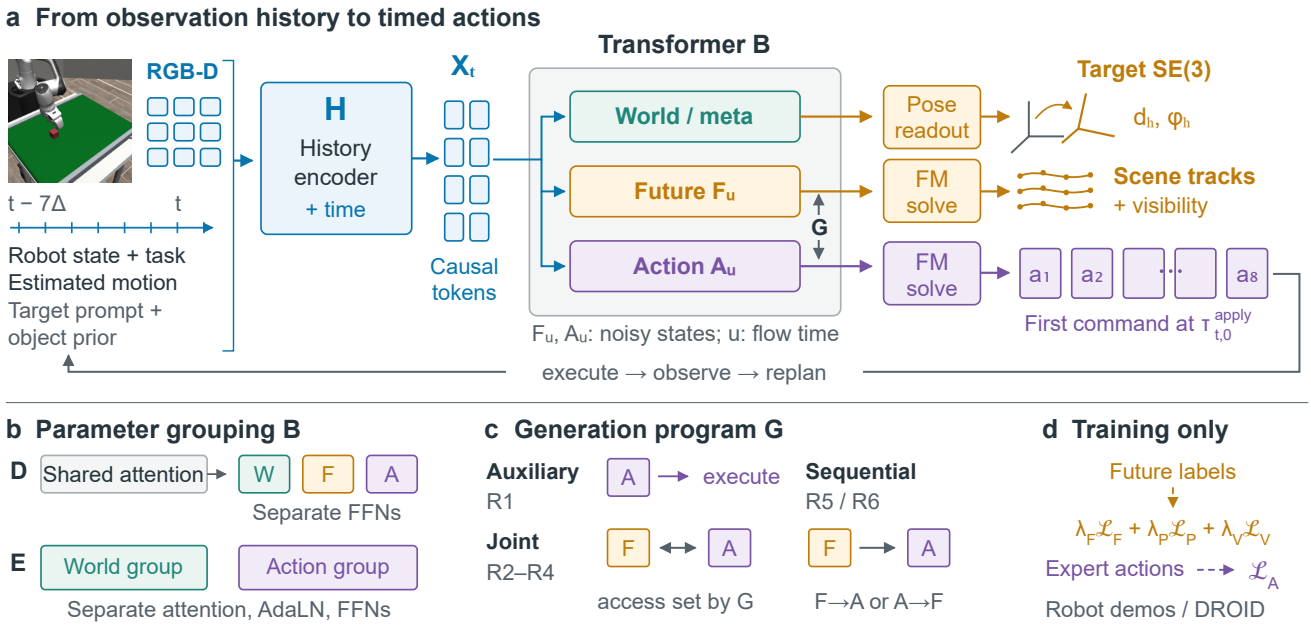



Figure 2. SE(3)-WAM: from timestamped history to interaction. The causal encoder supplies a conditioning bank to world and action streams. The world branch represents metric target pose and scene motion; the action branch generates a command chunk. Backbone grouping $B$ and information flow $G$ determine parameter sharing and access to generated world states. Future observations supervise training, while deployment uses observations available at the query time and generated states only. The lower panels separate auxiliary, joint, and sequential computation; Section 4 and Appendix E specify their training and sampling programs.

The visual-state reference uses masks from a frozen SAM 2 model [25] and FoundationPose RGB-D registration/tracking [31]. A target box in the first observed frame initializes the mask; object CAD or independent reference views are available before the episode. Tracking proceeds forward without future corrections. Invalid estimates produce an explicit missing-state token and validity bit, with the same interface during training. Appendix F specifies initialization, causal processing, and the shared input contract.

We compare current RGB, stacked RGB, codec motion vectors, optical flow, compressed video features, and age-dependent or uniform query compression. These alternatives expose different temporal information at different token costs. An oracle motion history supplies a reference for information quality; it is not a substitute for the visual deployment encoder. Q1 therefore separates current localization from motion access, and compares the same temporal support before attributing a difference to compression.

Target and context roles are derived from the task-conditioned current target assignment. The shared-scene alternative retains one token bank and learns task-dependent readout; explicit-role alternatives attach target/context embeddings. Static context remains observable in all cases. Role assignment at training time does not introduce future visibility into current tokens.

### 4.2. Future State and Prediction Resolution

The world representation contains target motion and a configurable scene description. For current target pose $(p_t, R_t)$ and offsets $\delta_h = 0.05h$ seconds, $h = 1, \ldots, 8$, the pose targets are

$$d_h = p_{t+\delta_h} - p_t, \qquad \phi_h = \mathrm{Log}(R_{t+\delta_h} R_t^\top)^\vee. \quad (3)$$

The reconstruction is $\hat{p}_{t+\delta_h} = p_t + \hat{d}_h$ and $\hat{R}_{t+\delta_h} = \mathrm{Exp}(\hat{\phi}_h^\wedge) R_t$. A deterministic readout of sixteen meta tokens predicts these cumulative translations and left-relative rotations. Its valid-target loss is squared translation error plus 0.1 times squared rotation geodesic error. The representation is metric; the network is not assumed equivariant.

Scene alternatives include metric point displacement, sparse image trajectories with visibility, and video features. Target alternatives include point trajectories or pooled visual features. These define Q2's variable comparison. A separate allocation comparison uses the same 128 scene tracks and eight future times: 64/64 target/context tracks, 96/32, or 32/96, with equal total tokens and identical role information. Target-only and context-only losses on the same support test prediction scope. Thus a change of spatial allocation can be separated from replacing pose with RGB or changing the number of tokens.

For online world generation, let $F_1$ be the standardized future state: $196 \times 8 \times 3$ scene displacement or $128 \times 8 \times 2$

track coordinates. Statistics use training data only. Track visibility is predicted by a separate binary head; ground-truth validity masks restrict losses and are not conditioning inputs. The pose readout remains deterministic. Appendices B and D define support, normalization, and the full target family. This shared interface allows the same future representation to serve as an auxiliary objective or an explicit action input.

### 4.3. World–Action Training and Generation

The reference transformer has four layers, width 768, and eight heads. The visual stem is held fixed within each comparison: a frozen DINOv3 encoder [26] or the jointly trained CNN used in the combination study. Backbone D shares attention projections and separates stream FFNs. Backbone E separates world and action parameter groups, each with its own attention, modulation, and FFN. Their active backbone capacities are matched within 5%; frontend and deployment parameters are reported separately.

Actions use flow matching [18]. For normalized target $a_1, a_0 \sim \mathcal{N}(0, I)$, and $u \sim \mathcal{U}[0, 1]$,

$$\begin{aligned} a_u &= (1-u)a_0 + ua_1, \\ \mathcal{L}_A &= \mathbb{E}\|v_\theta^A(a_u, u, X_t, \mathcal{C}_G) - (a_1 - a_0)\|^2. \end{aligned} \tag{4}$$

Here $\mathcal{C}_G$ is empty when Action does not read World. Otherwise it contains the noisy future and its flow time for joint generation, or the decoded first-stage state for R5. The generated world state uses the same interpolation with an independent Gaussian source. The total objective is $\mathcal{L} = \mathcal{L}_A + \lambda_F \mathcal{L}_F + \lambda_P \mathcal{L}_P + \lambda_V \mathcal{L}_V$, restricted to the enabled targets and valid support. Coefficients and normalization are shared within a contrast.

Q3 separates supervision, access, and sampling. R0 trains action alone. R1 adds future losses without world–action access. R2 lets action read world states; R3 lets world read actions; R4 permits both. These joint programs use five flow steps per generated modality. R5 first generates world and then action; R6 reverses that order, using five steps per stage. The second stage is trained with the clean demonstrated first-stage value and deployed with its generated value. This defines a sequential modeling choice, including its teacher-forcing mismatch, rather than an isolated change in solver order. Appendix E gives every attention mask and gradient path. R1's action-only/joint deployment pair uses the same checkpoint to isolate the cost of generating an unused forecast.

Each action chunk comprises eight E7 commands: base-frame translation increment, left-relative rotation vector, and gripper command. The controller executes the first command and replans. Conditioning is cached within a query; subsequent queries refresh the observation bank. Five Euler steps form the reference action solver. Execution uses physical timestamps from Eq. (1); measured latency is included in deployment tests.

### 4.4. Complete Model Selection and Robot Pretraining

Model selection operates on complete configurations, not independently chosen components. We revisit history under the retained future representations, future representation under alternative information flows, and D/E under the selected flow. The development rule compares task-balanced SR and p95 latency under a 200 ms latency ceiling and a memory ceiling of 80% of the common device's capacity. A two-percentage-point practical margin and a 10% latency reduction define the accuracy/cost tradeoff; unresolved contrasts retain the declared reference (Appendix E). Unused scenes confirm the selected design without reopening selection.

The resulting model is pretrained on DROID trajectories [14]. Causal pose/track estimates build inputs, while offline geometric estimates supply masked world labels. Calibration, temporal alignment, label error, and supervision coverage are evaluated independently. We compare action-only and world–action pretraining at matched data and exposure, then vary model capacity and unique data volume (Section 5.5).

A single robot-data base initializes simulation and real-robot adaptation independently. Each branch uses its own training data, normalization, and optimizer; neither inherits the other's adapted weights. Comparisons with no robot pretraining retain the same generic visual initialization. This construction tests whether a learned dynamics representation transfers, separately from the advantage of a downstream recipe.

## 5. Experiments

The evaluation addresses complete-system performance, the three design questions, and transfer from robot-data pretraining. Dashes denote unavailable measurements. Diagnostic results retain their original development partitions and information tracks. Appendix A describes scoring, and Appendices B–H provide complete configurations and results.

### 5.1. System Performance and Comparison Protocol

Table 1 compares action and world–action systems on Dynamic14, delay robustness, external benchmarks, and real manipulation. All final Dynamic14 scores use the unused 1,400-scene confirmation set and the visual deployment track. Each baseline retains its published model organization, with common observations, demonstrations, physical action semantics, and equal downstream tuning opportunities. Robot pretraining, trainable parameters, cumulative training exposure, and end-to-end cost are reported alongside performance. The scratch/pretrained SE(3)-WAM pair additionally shares downstream data and exposure, isolating initialization within the same model family.

Table 1. System evaluation with causal visual inputs. Dynamic14 reports task-macro success at zero and 150 ms added delay; all success columns are percentages. External suites use their native success definitions. Latency is the end-to-end sensor-to-command p95 on the common deployment device. Each row uses the same task demonstrations and evaluation scenes; initialization is part of the system comparison.

| | | Dynamic14 | | Transfer | | | Latency |
|---|---|---|---|---|---|---|---|
| System | Initialization | 0 ms | 150 ms | DOMINO | LIBERO | Real | p95 (ms) |
| Diffusion Policy [7] | Scratch | — | — | — | — | — | — |
| $\pi_{0.5}$ [24] | Published weights | — | — | — | — | — | — |
| DreamVLA [42] | Published recipe | — | — | — | — | — | — |
| SE(3)-WAM | Scratch | — | — | — | — | — | — |
| SE(3)-WAM | DROID pretraining | — | — | — | — | — | — |

Table 2. Single-seed history comparison with current GT target pose and robot history. GT motion adds privileged target history to current RGB. SR uses the 1,400-scene development total; $\Delta$ and its pointwise 95% scene-bootstrap interval are relative to current image, in percentage points. Appendix C gives the protocol, partition counts, and training differences.

| History input | SR (%) | $\Delta$ | 95% CI (pp) |
|---|---|---|---|
| Current image | 53.43 | 0.00 | — |
| Window | 47.29 | $-6.14$ | $[-8.43, -3.79]$ |
| Codec-MV | 52.36 | $-1.07$ | $[-3.36, 1.21]$ |
| Flow | 51.43 | $-2.00$ | $[-4.43, 0.43]$ |
| Flow12D | 51.64 | $-1.79$ | $[-4.21, 0.64]$ |
| PACT-Age | 45.29 | $-8.14$ | $[-10.64, -5.64]$ |
| Wan | 50.71 | $-2.71$ | $[-5.29, -0.14]$ |
| PACT-Uniform | 47.86 | $-5.57$ | $[-8.00, -3.14]$ |
| GT motion | 58.86 | $+5.43$ | $[3.07, 7.79]$ |

The earlier adapted-system comparison is retained in Appendix B with its original conditions. For example, DreamVLA's independent twenty- and forty-epoch recipes obtain 79.37% and 79.60% on the 1,260-scene development complement, whereas the selected privileged-information combination obtains 66.59%. Those scores measure complete adapted systems with different inputs and budgets; they are not transferred into the final visual-track table.

### 5.2. Q1: What History Is Useful for Control?

Q1 compares eight visual-history encodings with current target pose and robot history fixed, and adds a freshly trained GT-motion reference. Table 2 shows that Current reaches 53.43%, while none of the seven visual-history encodings exceeds that point estimate. GT motion reaches 58.86%, a 5.43-point increase, although phase passage loses 83 successes. The protocol tests the conditional value of history given localization, not recognition of the target from RGB alone.

Three complementary measurements explain the comparison. First, frozen readouts measure retained motion information: Window reduces past-position error at 350 ms from 10.89 to 8.46 mm. Second, repeating or reversing past images changes the actions of all seven history models, while both current-image references remain invariant. Third, closed-loop success measures utility: despite its improved readout, Window loses 6.14 SR points. Information retention and action sensitivity therefore do not imply better control under this recipe.

The frozen GT-motion intervention tests dependence on correct history. Replacing past poses with the current pose and recomputing kinematics reduces success from 824 to 9/1,400, a 58.21-point drop (95% paired interval 56.50–59.86). All 210 probe-window first actions change, with a mean translation change of 18.40 mm. An identity intervention preserves actions and reproduces all 140 control success/failure labels. This is an inference-time mismatch effect; the separately trained current-image model still reaches 53.43%.

The matched confirmation crosses temporal spacing and occlusion with visual and estimated-state histories. Repeating history holds token count constant; time reversal preserves appearance but changes motion order; age perturbation tests timing. Closed-loop intervention scores, contact errors, and encoder latency accompany motion readouts (Appendix C). Together these tests distinguish missing information, inaccurate encoding, and failure to use available motion.

### 5.3. Q2: What Future Should Be Predicted?

**Prediction variables.** The fifteen-condition screen fixes current RGB, DINO-L, target kinematics, training exposure, and action-only deployment. Four epochs correspond to 40,104 updates and 641,664 sampled chunks at global batch 16. No-future P00 obtains 791/1,260 successes; scene flow, target pose, and their combination obtain 782, 789, and 786. Sparse scene tracks with pose (P07) and flow with target tracks (P09) obtain 807 and 808. Their paired gains/losses are 85/69 and 91/74; neither shows a clear overall advantage.

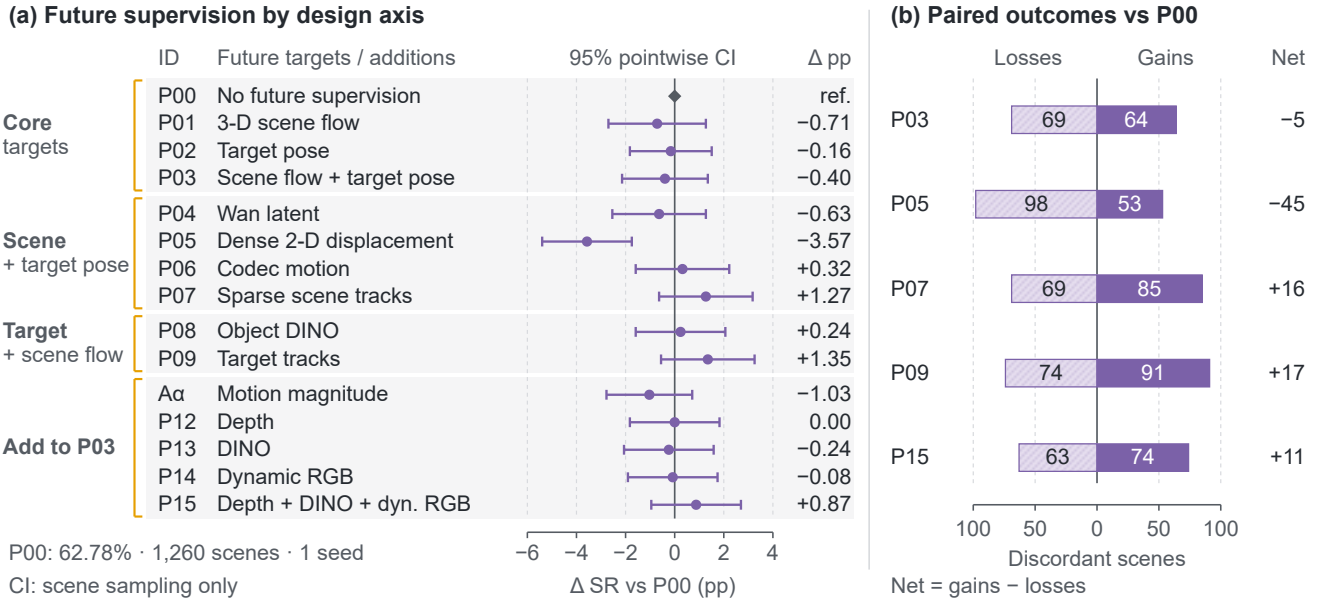



Figure 3. Future-supervision comparison on 1,260 paired development scenes: fourteen tasks, ninety each, one training seed. (a) SR differences from P00 (62.78%) and pointwise 95% task-stratified paired-bootstrap intervals for all fifteen conditions. (b) Scene-level gains and losses for selected contrasts. P05 is adverse after the twenty-seven-contrast correction; P07/P09 do not establish an overall gain. The fixed140 partition and per-task counts are in Appendix D.

Dense 2-D displacement with pose (P05) obtains 746/1,260, 3.57 points below P00 ($p_{\text{adjusted}} = 0.00844$). This adverse comparison concerns that representation package; the competitive sparse-track conditions also use 2-D coordinates.

**Representation interacts with the visual encoder.** A twelve-condition comparison crosses three future targets, DINO-S/CNN, and current-target reconstruction, retaining a full visual window and privileged motion. Scene tracks plus pose with CNN and reconstruction (candidate I) obtains 935/1,400 (66.79%). Matched no-future and flow/pose controls obtain 810 and 861. The gains are 8.93 and 5.29 points, respectively; 72 of the 74 net successes over flow/pose arise on phase passage. Candidate I is selected on this development pool, and its second training seed reaches 61.21%. Appendix D gives all combinations; the confirmation comparison repeats the selected condition and both controls on common training seeds and unused scenes.

**Scope and granularity.** The fifteen targets change variables and support jointly. We isolate allocation within the common track representation: target-only, context-only, uniform target/context, target-fine, and context-fine. All share current target identity, temporal support, and 128 tokens. The shared-bank control removes explicit role embeddings while retaining the same sampled points. This tests whether an explicit division of roles helps beyond task-dependent readout. Pose accuracy, visible-track error, contact error, and SR are reported together (Appendix D). A geometry error decrease without a task gain is interpreted as better prediction, not evidence for a superior control representation.

Table 3. Backbone and information flow. The upper contrast varies parameter grouping at fixed access. The lower block fixes the backbone $B^\star$ selected by the development rule and varies supervision, access, or generation order, with $B^\star$/R1 as its reference. SR is the three-seed mean task-macro success (%). End-to-end p95 includes the generated branches used by each program.

| Configuration | SR ↑ | p95 (ms) ↓ |
|---|---|---|
| D/R1: shared attention, split FFNs | — | — |
| E/R1: world/action grouping | — | — |
| $B^\star$/R0: action loss only | — | — |
| $B^\star$/R1: auxiliary, action only | — | — |
| $B^\star$/R2: joint, $F \to A$ | — | — |
| $B^\star$/R4: joint, $F \leftrightarrow A$ | — | — |
| $B^\star$/R5: future, then action | — | — |

## 5.4. Q3: How Should World and Action Interact?

Q3 first compares D/E with history, future targets, and auxiliary-only deployment fixed. It then fixes the backbone and varies the R0–R6 programs (Table 3; full matrix in Appendix E). The contrasts ask whether world supervision improves features (R0/R1), whether action benefits from

generated world state (R1/R2), whether action-conditioned world generation matters (R2/R4), and whether sequential generation changes the accuracy/cost tradeoff (R2/R5). Five joint steps and five-plus-five sequential steps match steps per modality, not total computation.

For models that consume world state, frozen-policy interventions replace the forecast with a matched zero-motion forecast, a shuffled forecast, or an oracle on its valid support. Pre-contact exogenous motion permits a direct oracle comparison; after contact, future labels are conditioned on the same candidate action in the simulator. We measure action displacement, contact timing, and paired SR. R1's action-only/joint comparison checks the cost of producing a forecast that action never reads. These measurements separate training effects, forecast dependence, and useful online anticipation.

### 5.5. Pretraining, Transfer, and Deployment

**Robot-data pretraining.** We compare no robot pretraining, action-only pretraining, and world–action pretraining, using a common downstream recipe and an equal-budget tuning comparison. Model capacities span approximately $0.5, 1, 2, 4$ times the reference backbone; nested data fractions are $1/8, 1/4, 1/2, 1$. Four full-data curves and three smaller subsets at two capacities give ten primary curves. A separate exposure comparison distinguishes additional unique experience from repeated training. Downstream demonstration fractions of 10/25/50/100% measure data efficiency. Appendix F specifies checkpoints, label-quality controls, learning curves, and the complete result matrix.

**Transfer beyond the benchmark.** Single-arm DOMINO measures external dynamic manipulation; LIBERO [19] measures compatibility with static multitask control. Real-robot evaluation covers moving grasping, moving-receptacle placement, timed passage, and post-impact interception. Each task crosses three speeds and three delay levels, using paired initial conditions. Every downstream branch starts independently from the same selected robot-data checkpoint. Domain-specific normalization and action-interface adaptations are reported with the native task scores (Appendix F).

**Latency and contact.** We evaluate imposed delays of 0/50/100/150/200 ms and a measured-latency execution mode. Timestamps cover capture, preprocessing, model inference, command dispatch, and command application. Prediction and action horizons are held in physical time. The diagnostic future-screen calls average 177.071–184.212 ms, with p95 178.726–185.133 ms, measured inside the policy boundary; those timings exclude sensor and command transport. The deployment table uses end-to-end latency. Delay-conditioned SR and contact errors reveal whether a predictor remains useful once its own computation makes the observation older.

**Failure analysis.** All fifteen future conditions fail rigid-motion receptacle placement; full constrained passage also remains near zero. A supervision audit finds release only at the last chunk position in all 800 full-passage demonstrations, whereas evaluation executes the first. The appendix separates this coverage issue from representation failures, and aligns qualitative forecasts, issued actions, and physical outcomes. These diagnostics identify where improved scene prediction alone is unlikely to resolve the control problem.

## 6. Conclusion

SE(3)-WAM studies dynamic manipulation through the information needed at interaction time. The model connects timestamped history, structured target dynamics, scene prediction, and action generation within explicit training and deployment interfaces. Its benchmark and experiments separate information quality, auxiliary supervision, online forecast use, and computation cost. The diagnostic evidence shows why these distinctions matter: a readable history need not improve control, and future-supervision benefits can depend strongly on the surrounding representation. The complete evaluation tests these design choices through matched confirmation, robot-data pretraining, and independent downstream adaptation.

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